EE3621: Digital Signal Processing

Lecture 5: Z-Transform, ROC & Common Pairs
(III B.Tech EEE)

Lecture Notes (40-Minute Plan)

1 Lecture Plan & Objectives (40 Minutes Breakdown)

- **00:00 – 08:00 (8 mins):** Motivation: Z-transform as a generalization of the DTFT ($z = re^{j\omega}$) and convergence proof.
- **08:00 – 16:00 (8 mins):** The Region of Convergence (ROC): Does ROC lie inside the unit circle? Causal vs. Anti-causal vs. Two-sided shapes and Stability Comments.
- **16:00 – 26:00 (10 mins):** Worked Examples: Exp, Sin, Cos, Unit Step, Impulse, and Multi-Pole Multi-Zero System (Magnitude & Phase Computation).
- **26:00 – 34:00 (8 mins):** Properties of the Z-Transform with full algebraic proofs.
- **34:00 – 38:00 (4 mins):** Comprehensive Common Z-Transform Pairs Table.
- **38:00 – 40:00 (2 mins):** Checkpoint and Concept Recap.

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2 Definition & Mathematical Motivation

The Z-transform of a discrete-time sequence $x[n]$ is defined as:

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} \tag{1}$$

Expressing z in polar form $z = re^{j\omega}$:

$$X(re^{j\omega}) = \sum_{n=-\infty}^{\infty} (x[n]r^{-n})e^{-j\omega n} \tag{2}$$

Evaluating $X(z)$ on the unit circle ($|z| = 1 \implies r = 1$) yields the Discrete-Time Fourier Transform (DTFT): $X(z)\Big|_{|z|=1} = X(e^{j\omega})$. The Z-transform series converges absolutely if:

$$\sum_{n=-\infty}^{\infty} |x[n]r^{-n}| < \infty \tag{3}$$

The set of complex values z for which this summation converges defines the **Region of Convergence (ROC)**.

3 Properties of the ROC & Unit Circle Location Clarification

3.1 Does the ROC Lie Inside the Unit Circle?

Whether an ROC lies inside the unit circle depends directly on signal causality and pole magnitude:

1. **Anti-Causal (Left-Sided) Signals:** The ROC is an **interior disk** $|z| < |p|_{min}$. If $|p| < 1$ (e.g., $p = 0.7$), the ROC is $|z| < 0.7$, which **lies entirely INSIDE the unit circle** ($|z| < 1$).
2. **Causal (Right-Sided) Signals:** The ROC is an **exterior domain** $|z| > |p|_{max}$ extending to infinity.
3. **Two-Sided Signals:** The ROC is an **annular ring** $r_1 < |z| < r_2$.
4. **BIBO Stability Criterion:** An LTI system is BIBO stable if and only if the ROC **contains the unit circle** ($|z| = 1$).

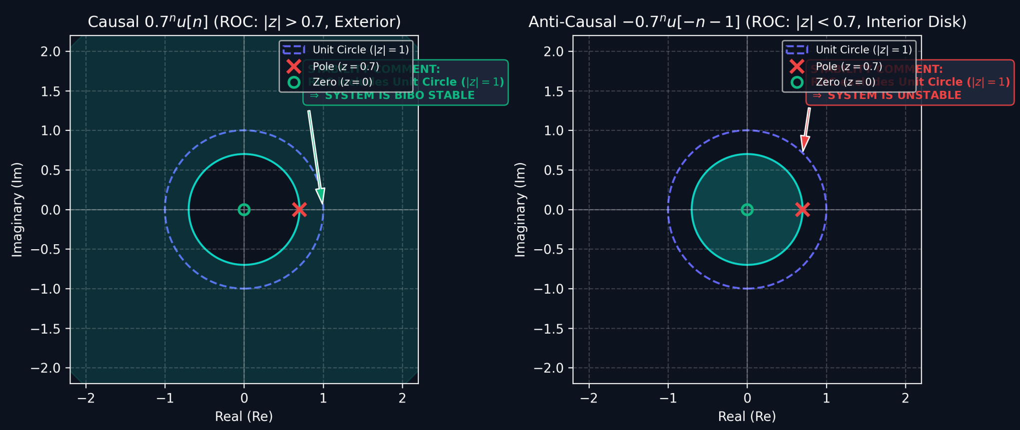

Figure 1: ROC geometries in the z-plane for causal ($|z| > 0.7$) vs. anti-causal ($|z| < 0.7$, inside unit circle) signals with explicit BIBO stability comments.

4 Worked Examples: Signal Variants & Multi-Pole Multi-Zero Systems

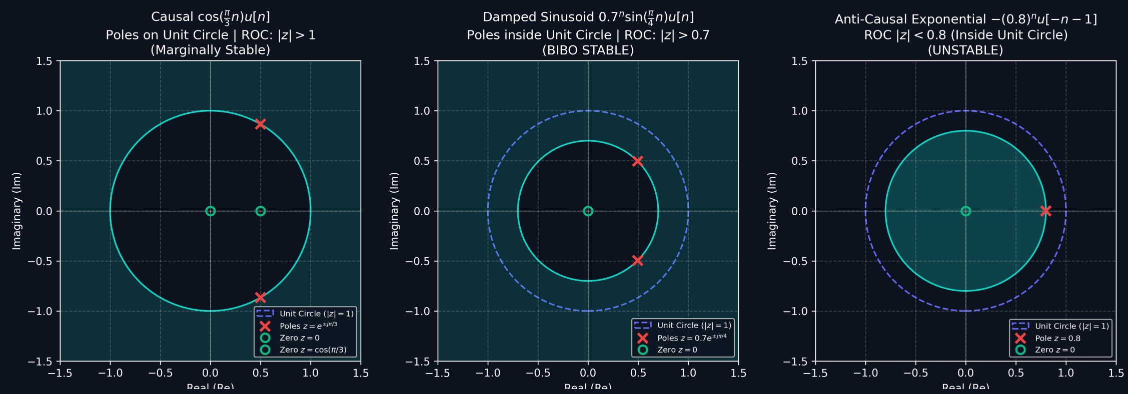

Figure 2: Demonstrative Pole-Zero plots and ROC regions for Causal Cosine, Damped Sinusoid, and Anti-Causal Exponential.

4.1 Example 1: Impulse, Step, Cosine & Damped Sinusoid

- **Unit Impulse $\delta[n]$:** $X(z) = 1$, ROC: All z. **Stable**.
- **Unit Step $u[n]$:** $X(z) = \frac{z}{z-1}$, Pole at $z = 1$, ROC: $|z| > 1$. **Marginally Stable**.
- **Cosine $\cos(\omega_0 n)u[n]$:** $X(z) = \frac{z(z-\cos\omega_0)}{z^2-2\cos\omega_0 z+1}$, Poles at $z = e^{\pm j\omega_0}$ (on unit circle), ROC: $|z| > 1$. **Marginally Stable**.
- **Damped Sinusoid $r_0^n \sin(\omega_0 n)u[n]$ ($r_0 = 0.7, \omega_0 = \pi/4$):** $X(z) = \frac{0.7\sin(\pi/4)z}{z^2-1.4\cos(\pi/4)z+0.49}$, Poles at $z = 0.7e^{\pm j\pi/4}$, ROC: $|z| > 0.7$. **BIBO Stable**.

4.2 Example 2: Multi-Pole Multi-Zero System (ROC, Magnitude & Phase)

Consider $H(z) = \frac{z-0.5}{z^2-0.8z+0.25}$.

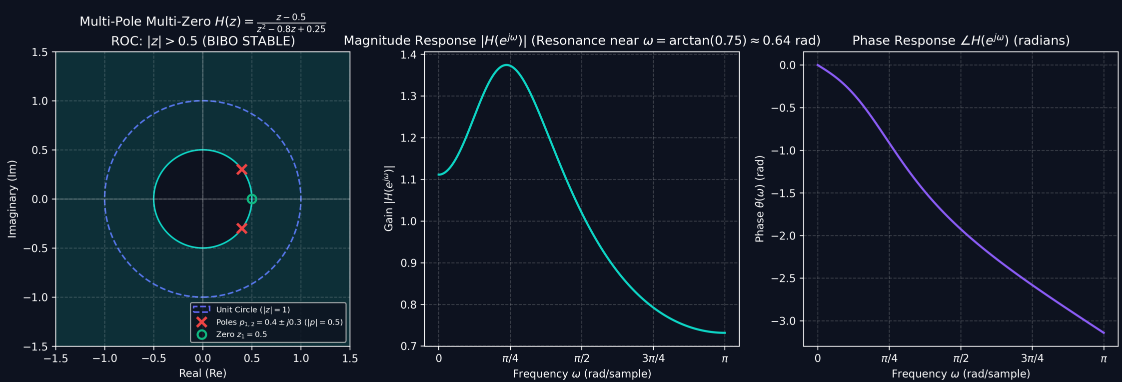

Figure 3: Multi-pole multi-zero system pole-zero plot in z-plane (left), magnitude response $|H(e^{j\omega})|$ (center), and phase response $\angle H(e^{j\omega})$ (right).

1. **Poles & Zeros:** Zero at $z_1 = 0.5$; Complex poles at $p_{1,2} = 0.4 \pm j0.3$ ($|p| = 0.5$).
2. **ROC & Stability:** Causal $\implies$ $|\mathbf{z}| > \mathbf{0.5}$. Contains unit circle $|z| = 1$ $\implies$ **BIBO STABLE**.

3. **Magnitude Computation $|H(e^{j\omega})|$:**

$$|H(e^{j\omega})| = \frac{|e^{j\omega} - 0.5|}{|e^{j\omega} - (0.4 + j0.3)| \cdot |e^{j\omega} - (0.4 - j0.3)|} \tag{4}$$

At DC ($\omega = 0$): $|H(e^{j0})| = \frac{0.5}{0.45} = 1.111$ (1.09 dB). At resonance ($\omega \approx 0.6435$ rad): $|H(e^{j0.6435})| \approx 1.54$ (3.75 dB). At $\omega = \pi$: $|H(e^{j\pi})| = \frac{1.5}{2.05} = 0.7317$ (-2.71 dB).

4. **Phase Computation $\theta(\omega)$:**

$$\theta(\omega) = \angle(e^{j\omega} - 0.5) - \angle(e^{j\omega} - (0.4 + j0.3)) - \angle(e^{j\omega} - (0.4 - j0.3)) \tag{5}$$

5 Geometric Vector Method: Magnitude & Phase Computation

Evaluating frequency response by moving along the unit circle $z = e^{j\omega}$:

$$|H(e^{j\omega})| = |C| \cdot \frac{\prod_{k=1}^{M} |\vec{v}_{zk}(\omega)|}{\prod_{m=1}^{N} |\vec{v}_{pm}(\omega)|} \tag{6}$$

$$\angle H(e^{j\omega}) = \angle C + \sum_{k=1}^{M} \angle \vec{v}_{zk}(\omega) - \sum_{m=1}^{N} \angle \vec{v}_{pm}(\omega) \tag{7}$$

5.1 Case 1: Single Zero System ($H(z) = z - 0.6$)

$|H(e^{j\omega})| = |\vec{v}_{z1}| = \sqrt{1.36 - 1.2\cos\omega}$, $\angle H(e^{j\omega}) = \text{atan2}(\sin\omega, \cos\omega - 0.6)$. Always **BIBO Stable**.

Frequency ω	**Complex z**	**Zero Vector $\vec{v}_{z1}$**	**Magnitude $\|H(e^{j\omega})\|$**	**Phase $\angle H(e^{j\omega})$**
$\omega = 0$	$1 + j0$	$0.4 + j0$	0.4000 (Min)	$0.00°$ (0.000 **rad**)
$\omega = \pi/2$	$0 + j1$	$-0.6 + j1.0$	1.1662	$120.96°$ (2.111 **rad**)
$\omega = \pi$	$-1 + j0$	$-1.6 + j0$	1.6000 (Max)	$180.00°$ (π **rad**)
$\omega = 3\pi/2$	$0 - j1$	$-0.6 - j1.0$	1.1662	$-120.96°$ (-2.111 **rad**)
$\omega = 2\pi$	$1 + j0$	$0.4 + j0$	0.4000 (Min)	$0.00°$ (0.000 **rad**)

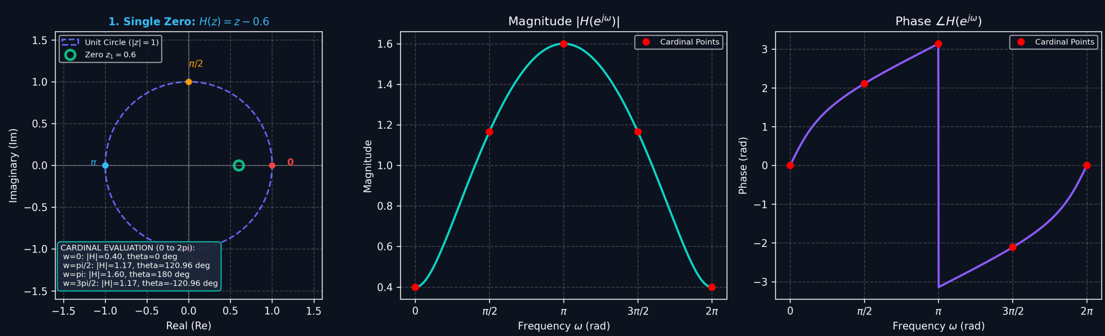

Figure 4: Single zero geometric vector evaluation along the unit circle from $0 \to 2\pi$.

5.2 Case 2: Single Pole System ($H(z) = \frac{1}{z-0.7}$)

$|H(e^{j\omega})| = \frac{1}{|\vec{v}_{p1}|} = \frac{1}{\sqrt{1.49 - 1.4\cos\omega}}$. **BIBO Stable** ($|0.7| < 1$).

Frequency ω	**Complex** z	**Pole Vector** $\vec{v}_{p1}$	**Magnitude** $\lvert H(e^{j\omega})\rvert$	**Phase** $\angle H(e^{j\omega})$
$\omega = 0$	$1 + j0$	$0.3 + j0$	3.3333 (Peak)	$0.00°$ **(0.000 rad)**
$\omega = \pi/2$	$0 + j1$	$-0.7 + j1.0$	0.8192	$-125.00°$ **(-2.182 rad)**
$\omega = \pi$	$-1 + j0$	$-1.7 + j0$	0.5882 (Min)	$-180.00°$ **($-\pi$ rad)**
$\omega = 3\pi/2$	$0 - j1$	$-0.7 - j1.0$	0.8192	$+125.00°$ **($+2.182$ rad)**
$\omega = 2\pi$	$1 + j0$	$0.3 + j0$	3.3333 (Peak)	$0.00°$ **(0.000 rad)**

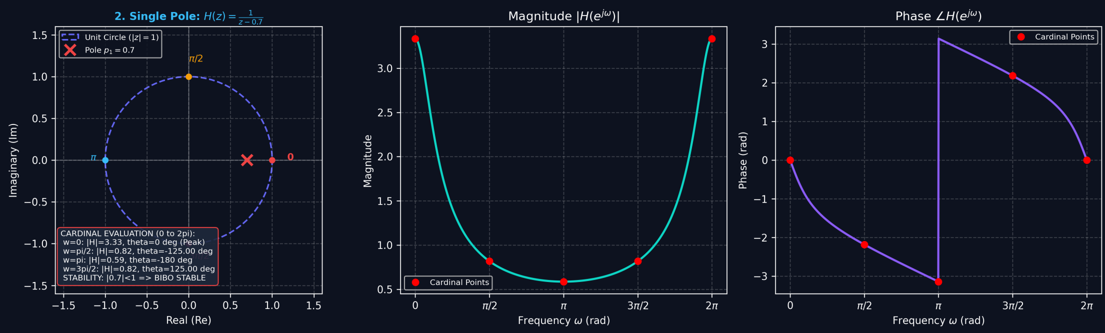

Figure 5: Single pole geometric vector evaluation showing resonance peak at $\omega = 0$.

5.3 Case 3: One Pole + One Zero System ($H(z) = \frac{z-0.8}{z-0.5}$)

$|H(e^{j\omega})| = \frac{|\vec{v}_{z1}|}{|\vec{v}_{p1}|} = \sqrt{\frac{1.64-1.6\cos\omega}{1.25-1.0\cos\omega}}$, $\angle H = \theta_{z1} - \phi_{p1}$. **BIBO Stable** ($|0.5| < 1$).

Frequency ω	**Zero Vector** $\vec{v}_{z1}$	**Pole Vector** $\vec{v}_{p1}$	**Magnitude** $\lvert H(e^{j\omega})\rvert$	**Phase** $\angle H(e^{j\omega})$
$\omega = 0$	$0.2 + j0$	$0.5 + j0$	0.4000	$0.00°$ **(0.000 rad)**
$\omega = \pi/2$	$-0.8 + j1.0$	$-0.5 + j1.0$	1.1454	$+12.09°$ **($+0.211$ rad)**
$\omega = \pi$	$-1.8 + j0$	$-1.5 + j0$	1.2000	$0.00°$ **(0.000 rad)**
$\omega = 3\pi/2$	$-0.8 - j1.0$	$-0.5 - j1.0$	1.1454	$-12.09°$ **(-0.211 rad)**
$\omega = 2\pi$	$0.2 + j0$	$0.5 + j0$	0.4000	$0.00°$ **(0.000 rad)**

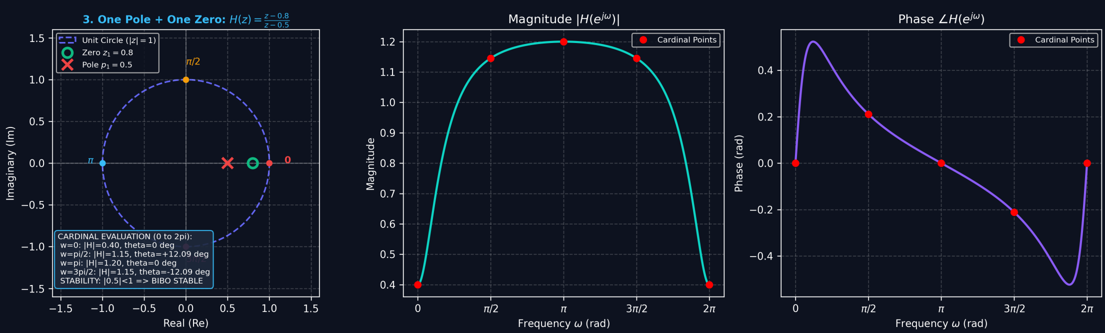

Figure 6: One pole and one zero geometric vector evaluation along unit circle.

5.4 Case 4: Two Poles + Two Zeros System ($H(z) = \frac{z^2-1}{z^2-0.8z+0.25}$)

Zeros at $z = \pm 1$ (notches at $\omega = 0, \pi$), complex poles at $p_{1,2} = 0.4 \pm j0.3$ ($|p| = 0.5$, peak at $\omega \approx 0.6435$ rad). **BIBO Stable**.

Frequency ω	**Numerator** $(e^{j2\omega}-1)$	**Denominator**	**Magnitude** $\lvert H\rvert$	**Phase** $\angle H(e^{j\omega})$
$\omega = 0$	0	0.45	0.0000 (Notch)	0.00°
$\omega = \pi/2$	-2	$-0.75 - j0.8$	1.8239	$+133.15^\circ$ **(+2.324 rad)**
$\omega = \pi$	0	2.05	0.0000 (Notch)	0.00°
$\omega = 3\pi/2$	-2	$-0.75 + j0.8$	1.8239	-133.15° **(−2.324 rad)**
$\omega = 2\pi$	0	0.45	0.0000 (Notch)	0.00°

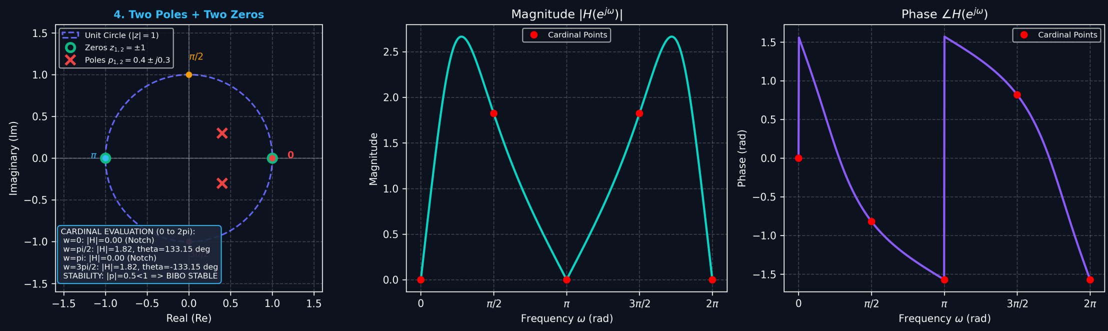

Figure 7: Two poles and two zeros geometric vector evaluation with notches and resonance.

6 Z-Transform Properties & Proofs

For $x[n] \leftrightarrow X(z)$ (ROC R_x) and $y[n] \leftrightarrow Y(z)$ (ROC R_y):

6.1 Time Shifting

$$x[n-n_0] \longleftrightarrow z^{-n_0}X(z) \tag{8}$$

6.2 Scaling in the Z-domain

$$a^n x[n] \longleftrightarrow X(a^{-1}z) \tag{9}$$

6.3 Differentiation in the Z-domain

$$nx[n] \longleftrightarrow -z\frac{dX(z)}{dz} \tag{10}$$

7 Comprehensive Common Z-Transform Pairs Table

Table 1: Comprehensive Z-transform pairs with ROC and Stability comments.

Signal $x[n]$	**Z-Transform** $X(z)$	**ROC**	**Poles / Zeros**	**Stability**
$\delta[n]$	1	All z	None	**Stable**
$u[n]$	$\frac{z}{z-1}$	$\lvert z\rvert > 1$	Pole: $z = 1$, Zero: $z = 0$	**Marginally Stable**
$a^n u[n]$	$\frac{z}{z-a}$	$\lvert z\rvert > \lvert a\rvert$	Pole: $z = a$, Zero: $z = 0$	**Stable** ($\lvert a\rvert < 1$)
$-a^n u[-n-1]$	$\frac{z}{z-a}$	$\lvert z\rvert < \lvert a\rvert$	Pole: $z = a$, Zero: $z = 0$	**Stable** ($\lvert a\rvert > 1$)
$nu[n]$	$\frac{z}{(z-1)^2}$	$\lvert z\rvert > 1$	Double Pole: $z = 1$	**Unstable**
$na^n u[n]$	$\frac{az}{(z-a)^2}$	$\lvert z\rvert > \lvert a\rvert$	Double Pole: $z = a$	**Stable** ($\lvert a\rvert < 1$)
$\cos(\omega_0 n)u[n]$	$\frac{z(z-\cos\omega_0)}{z^2-2\cos\omega_0 z+1}$	$\lvert z\rvert > 1$	Poles: $z = e^{\pm j\omega_0}$	**Marginally Stable**
$\sin(\omega_0 n)u[n]$	$\frac{z\sin\omega_0}{z^2-2\cos\omega_0 z+1}$	$\lvert z\rvert > 1$	Poles: $z = e^{\pm j\omega_0}$	**Marginally Stable**
$r^n\cos(\omega_0 n)u[n]$	$\frac{z(z-r\cos\omega_0)}{z^2-2r\cos\omega_0 z+r^2}$	$\lvert z\rvert > r$	Poles: $z = re^{\pm j\omega_0}$	**Stable** ($r < 1$)
$r^n\sin(\omega_0 n)u[n]$	$\frac{zr\sin\omega_0}{z^2-2r\cos\omega_0 z+r^2}$	$\lvert z\rvert > r$	Poles: $z = re^{\pm j\omega_0}$	**Stable** ($r < 1$)

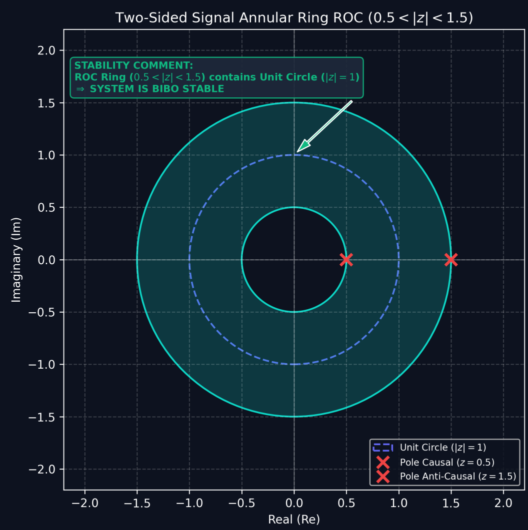

Figure 8: ROC of a stable two-sided sequence, forming an annular ring bounded by pole radii.